Pedagogical Support Framework

Youth in AI × Proficio Global Impact Foundation

Introduction

The success of the Youth in AI × Proficio Global Impact Foundation programme depends not only on the quality of its technical curriculum but also on the effectiveness of its teaching. As the **Learning Design and Education Lead**, the Pedagogical Support Framework is designed to equip facilitators with the knowledge, strategies, and confidence needed to deliver engaging, learner-centred, and inclusive instruction.

The framework ensures that every facilitator adopts evidence-based teaching practices that promote deep understanding, active participation, practical application, and continuous improvement. It also establishes common standards for instructional quality, learner engagement, assessment, and professional development, ensuring consistency across all training cohorts. This aligns with the role's responsibility to provide pedagogical support, develop engagement strategies for virtual learning, and support facilitators in instructional delivery.

Objectives of the Pedagogical Support Framework

The framework aims to:

- Equip facilitators with effective teaching strategies for technical education.
- Promote learner-centred and competency-based instruction.
- Ensure consistency in lesson delivery across facilitators.
- Support facilitators in managing diverse learning needs.
- Enhance participant engagement in both virtual and practical learning environments.
- Foster reflective teaching and continuous professional growth.
- Improve learning outcomes through evidence-based instructional practices.

Pedagogical Philosophy

The programme adopts a learner-centred philosophy grounded in constructivist and experiential learning principles. Participants are expected to actively construct knowledge by engaging with authentic problems, collaborating with peers, and reflecting on their learning experiences.

The instructional approach is guided by the following principles:

- Active participation over passive listening.
- Learning by doing through practical application.
- Collaboration and peer learning.
- Continuous feedback and improvement.
- Inclusive teaching that accommodates diverse learners.
- Reflection as a tool for deeper understanding.
- Real-world relevance in every learning activity.

Core Pedagogical Approaches

1. Learner-Centred Instruction

Facilitators serve as learning coaches rather than lecturers. Their role is to guide inquiry, encourage participation, and support learners as they build understanding.

Facilitators should:

- encourage questions and curiosity
- build on participants' prior knowledge
- adapt instruction to learners' needs
- promote independent thinking
- create opportunities for learner voice and choice

Expected outcome:

Participants take ownership of their learning and develop confidence in solving problems independently.

2. Active Learning

Every lesson should require participants to actively engage with the content rather than simply receive information.

Strategies include:

- live coding
- collaborative problem-solving
- think-pair-share
- brainstorming
- polling
- quizzes
- debugging exercises
- coding sprints
- practical demonstrations

Target:

At least **70% of every instructional session** should involve participant activity rather than facilitator talk.

3. Project-Based Learning

Participants learn best when solving authentic problems.

Each phase concludes with a practical project that integrates the concepts learned.

Projects should:

- address real-world Ghanaian challenges
- encourage collaboration
- require research and critical thinking
- produce meaningful outputs
- culminate in presentations and demonstrations

Examples include:

- analysing educational performance data
- building a dashboard for local health data
- developing an AI chatbot for student support

- creating predictive models for agricultural planning

4. Inquiry-Based Learning

Rather than immediately providing answers, facilitators encourage learners to investigate questions, analyse evidence, and draw conclusions.

Facilitators should:

- ask open-ended questions
- encourage experimentation
- promote hypothesis testing
- guide learners through discovery
- facilitate reflective discussions

5. Collaborative Learning

AI development is inherently collaborative, and instruction should mirror professional practice.

Collaborative activities include:

- pair programming
- peer code reviews
- group research
- team-based capstone projects
- collaborative presentations
- discussion forums

Expected outcome:

Participants develop communication, teamwork, and leadership skills alongside technical competence.

6. Experiential Learning

Participants learn by applying theory to practice through a continuous cycle of experience, reflection, conceptualisation, and experimentation.

Examples include:

- analysing real datasets
- deploying machine learning models
- testing AI tools
- conducting field-based data collection where appropriate
- presenting findings to peers

Facilitator Support System

Pre-Programme Training

Before facilitating, all trainers will participate in a structured orientation covering:

Programme Orientation

- programme goals
- curriculum overview
- learner profiles
- expected standards

Teaching Methodologies

- learner-centred teaching
- active learning techniques
- facilitation skills
- questioning strategies
- classroom management
- differentiated instruction

Educational Technology

Facilitators will receive training on:

- Google Classroom
- Google Colab
- Jupyter Notebook
- Zoom or Microsoft Teams
- Power BI
- collaborative digital tools
- assessment platforms

Facilitator Handbook

Each facilitator will receive a comprehensive handbook containing:

- programme philosophy
- instructional expectations
- lesson plans
- teaching notes
- timing guides
- engagement strategies
- assessment guidance
- troubleshooting tips
- frequently asked questions
- observation rubrics

Micro-Teaching Programme

Before delivering live sessions, facilitators will complete micro-teaching exercises.

Each facilitator will:

- prepare a 20-minute lesson
- teach peers
- receive structured feedback
- reflect on their performance
- revise their instructional approach

Assessment criteria include:

- clarity of explanation
- learner engagement
- pacing
- questioning techniques
- technical accuracy
- communication skills

Coaching and Mentoring

Facilitators will receive continuous support through:

Weekly Reflection Meetings

Discussions focus on:

- successes
- learner challenges
- instructional adjustments
- resource needs

Peer Observation

Facilitators observe one another and exchange constructive feedback using standard observation tools.

Instructional Coaching

The Educational Lead conducts coaching sessions addressing:

- classroom management
- learner engagement
- differentiation
- assessment practices
- reflective teaching

Supporting Diverse Learners

Recognising that participants have varying educational backgrounds, facilitators will implement differentiated instruction.

Differentiation Strategies

Content

Provide:

- introductory readings
- extension materials
- optional advanced challenges

Process

Offer:

- guided practice
- independent learning
- peer tutoring
- collaborative learning

Product

Allow learners to demonstrate understanding through:

- coding assignments
- presentations
- dashboards
- written reflections
- demonstrations

Virtual Teaching Guidelines

Because much of the programme will be delivered online, facilitators should:

- maintain frequent interaction
- encourage camera use where feasible
- use breakout rooms effectively
- conduct regular knowledge checks
- minimise lecture time
- incorporate collaborative digital tools
- provide timely feedback
- record sessions for revision when appropriate

Learner Engagement Strategies

Facilitators will incorporate:

- weekly coding competitions
- discussion forums
- live polls
- AI news discussions
- guest speaker sessions
- hackathons
- capstone showcases
- digital badges
- recognition awards
- peer mentoring

Feedback Practices

Facilitators will provide feedback that is:

Timely

Within one week of task submission.

Constructive

Specific guidance highlighting strengths and areas for improvement.

Actionable

Clear next steps that learners can implement immediately.

Balanced

Recognising achievement while encouraging continued growth.

Assessment Support

Facilitators will receive guidance on:

- designing formative assessments
- using assessment rubrics
- providing meaningful feedback
- maintaining assessment consistency
- identifying learners requiring additional support

Inclusive Teaching Practices

Facilitators are expected to:

- use accessible language
- avoid unnecessary technical jargon
- explain concepts using multiple examples
- encourage participation from all learners
- respect cultural diversity
- create psychologically safe learning environments
- accommodate different learning speeds

Quality Assurance

To maintain instructional excellence, the Educational Lead will implement a quality assurance process that includes:

Classroom Observation

Each facilitator will be observed at least twice during every training phase.

Observation areas include:

- lesson organisation
- learner engagement
- instructional clarity
- pacing
- questioning techniques
- technology integration
- assessment practices

Learner Feedback

Participants will complete anonymous evaluations after each module, providing insights into facilitation quality, clarity, engagement, and support.

Teaching Portfolio

Facilitators will maintain a professional portfolio containing:

- lesson reflections
- observation reports
- learner feedback summaries
- professional development records
- examples of teaching resources

Continuous Professional Development

Facilitators will participate in ongoing professional learning through:

- monthly pedagogical workshops
- webinars on AI education
- instructional design seminars
- peer learning communities
- educational research discussions
- technology update sessions
- reflective practice groups

Expected Outcomes

Implementation of this Pedagogical Support Framework will ensure that facilitators:

- Deliver consistently high-quality, learner-centred instruction across all cohorts.
- Confidently teach complex AI and data science concepts using evidence-based pedagogical approaches.
- Foster active participation, collaboration, and critical thinking in every learning session.
- Effectively support learners with diverse educational backgrounds and learning needs.
- Integrate formative assessment and constructive feedback to enhance learner progress.
- Continuously improve their instructional practice through coaching, reflection, observation, and professional development.
- Contribute to a sustainable community of skilled AI educators capable of expanding the programme through the Train-the-Trainer model.

Ultimately, this framework transforms facilitators from subject matter experts into effective educators, ensuring that technical excellence is matched by equally strong teaching practice and meaningful learner outcomes.